

Inverse Inversions

Time limit: 1.00 s

Memory limit: 512 MB

Your task is to create a permutation of numbers $1, 2, \dots, n$ that has exactly k inversions.

An inversion is a pair (a, b) where $a < b$ and $p_a > p_b$ where p_i denotes the number at position i in the permutation.

Input

The only input line has two integers n and k .

Output

Print a line that contains the permutation. You can print any valid solution.

Constraints

- $1 \leq n \leq 10^6$
- $0 \leq k \leq \frac{n(n-1)}{2}$

Example

Input	Output
5 4	1 5 2 4 3